

From GANs to Diffusion and Drifting

Generative Channel Models for 6G

Rafael F. Schaefer Rick Fritschek

Chair of Information Theory and Machine Learning, TUD Dresden University of Technology

Berlin 6G Conference



How should a learned channel balance
fidelity, conditional structure, and simulation cost?

FIDELITY

CONDITIONING

DIFFERENTIABLE

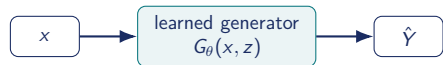
FAST SAMPLING

A learned channel is a conditional sampler



$$Y \sim p_{\text{ch}}(y | x)$$

$$Z \sim p_Z$$

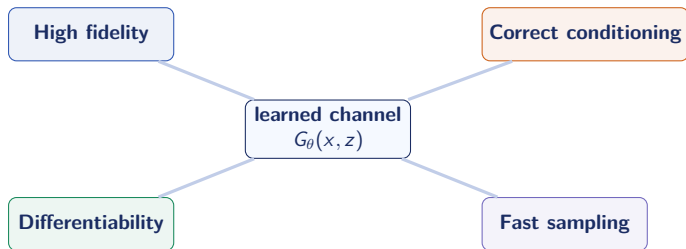


The target

$$p_{G_\theta(X,Z)|X}(\cdot | x) \approx p_{\text{ch}}(\cdot | x)$$

- ▶ nonlinear and mismatched hardware
- ▶ data-driven channel characterization
- ▶ differentiable end-to-end learning
- ▶ repeated Monte Carlo simulation

A useful simulator must satisfy four constraints



Match distribution shape, variability, and relevant tails.

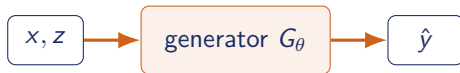
Preserve the output law associated with each transmitted input.

Allow optimization through the learned channel response.

Keep repeated training and Monte Carlo calls affordable.

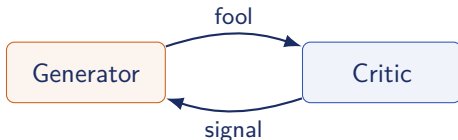
GANs offer direct samples; WGAN is our baseline

INFERENCE



Direct conditional generation
one generator evaluation

TRAINING



- ▶ adversarial min–max optimization
- ▶ stability and balance
- ▶ distribution coverage

I. Goodfellow et al., NeurIPS 2014. M. Arjovsky et al., ICML 2017. The experiments in this talk use a conditional WGAN baseline.

GAN channel models established the differentiable-surrogate idea

Unknown conditional channels

Surrogates for end-to-end transmitter and receiver learning.

H. Ye et al., GLOBECOM Wkshps. 2018

H. Ye et al., IEEE TWC 2020

Richer channels

Time variation and frequency-selective responses.

Y. Yang et al., IEEE Commun. Mag. 2019

D. Singh and A. K. R. Chavva, IEEE ICC 2022

MIMO and twins

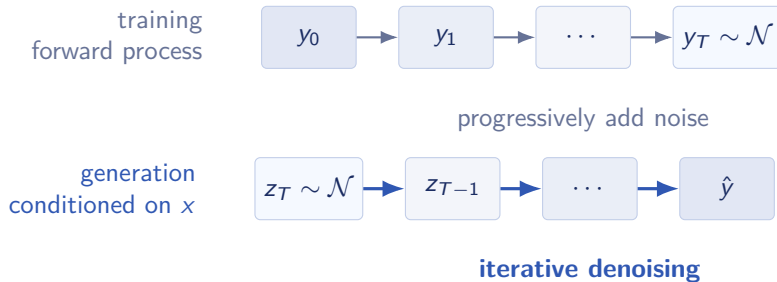
Impulse responses, spatial statistics, and data-driven channel twins.

T. Orekondy et al., IEEE ICC 2022

Y. Zhang et al., JGIN 2023

conditional surrogate → **channel memory** → **spatial structure**

Diffusion learns difficult channel laws through denoising



STRENGTH

strong distributional modeling

COST

sequential generation

J. Ho et al., NeurIPS 2020. J. Song et al., ICLR 2021. M. Kim et al., IEEE ICC 2024.

Diffusion reset the fidelity benchmark for image generation

2020: DDPM

competitive high-quality image synthesis



2021: guided diffusion

surpassed leading GAN benchmarks with better coverage



2022: latent diffusion

scaled flexible conditioning to high-resolution synthesis

Why it became central

- ▶ strong sample fidelity
- ▶ broad distribution coverage
- ▶ flexible conditioning
- ▶ no adversarial min-max game

The remaining cost is **iterative generation.**

J. Ho et al., NeurIPS 2020. P. Dhariwal and A. Nichol, NeurIPS 2021. R. Rombach et al., CVPR 2022.

“a photograph of
an astronaut
riding a horse”



Stable Diffusion 3.5
VulcanSphere
Wikimedia Commons
CC0

Diffusion is now used across the wireless stack

OUR WORK

Channel generation

Conditional sampling, end-to-end learning, and reduced-step robustness.

Kim, Fritschek, Schaefer, accepted, IEEE TMLCN 2026

Kim, Fritschek, Schaefer, IEEE ICC 2024

Channel synthesis

User-specific channels and sensing-assisted statistical CSI.

T. Lee et al., IEEE TWC 2026

X. Gong et al., IEEE TWC 2025

Estimation & detection

MIMO priors, high-dimensional estimation, joint detection, and time-varying MIMO-OFDM.

B. Fesl et al., IEEE WCL 2024

X. Zhou et al., IEEE TWC 2025

N. Zilberstein et al., IEEE ICASSP 2024

H. Fu et al., DSP 2025

The growing application range makes sampling cost increasingly relevant.

Iterative sampling turns channel calls into system cost

One-shot generator

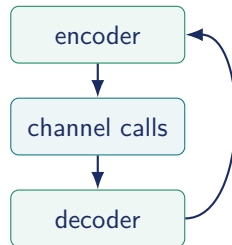


Diffusion generator



Where the cost appears

- ▶ every training mini-batch
- ▶ every Monte Carlo sample
- ▶ every differentiable channel call



**Can we obtain direct noise-to-channel generation
without adversarial training and without iterative
denoising?**

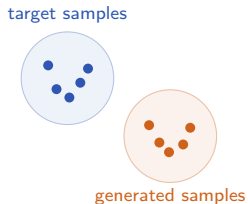


distributional dynamics during training

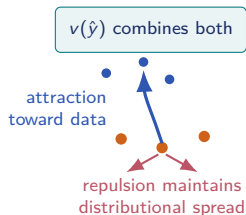
Drifting investigates this operating point.

Drifting learns a direct generator from particle directions

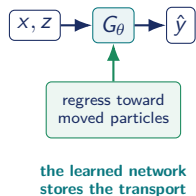
1. Compare particles



2. Construct a direction



3. Train the map



The field is used to train G_θ . Sampling remains one direct network evaluation.

M. Deng et al., "Generative Modeling via Drifting," arXiv:2602.04770, 2026.

A detached drift target trains the direct map

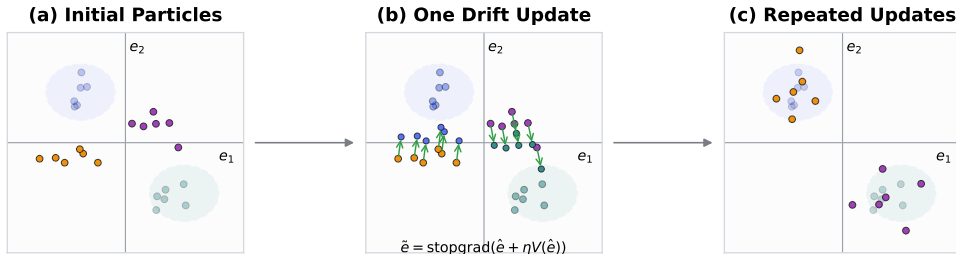


$$\tilde{y} = \text{stopgrad}(\hat{y} + \eta v(\hat{y})), \quad \min_{\theta} \|G_{\theta}(x, z) - \tilde{y}\|_2^2$$

CORE DESIGN CHOICE

The field v determines what distributional structure is learned.

Repeated drift training reshapes conditional output clouds



● target, $x^{(1)}$ ● target, $x^{(2)}$ ● generated, $x^{(1)}$ ● generated, $x^{(2)}$

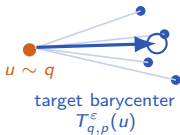
training: targets from a distributional field

inference: one network evaluation

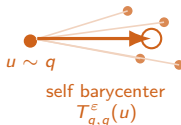
Illustration follows the conditional drifting construction used in this work.

W-Flow turns Sinkhorn couplings into particle directions

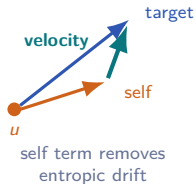
1. Couple to the target



2. Couple to itself



3. Subtract the bias



$$V_{\epsilon}(u) = T_{q,p}^{\epsilon}(u) - T_{q,q}^{\epsilon}(u)$$

Use $u + \eta V_{\epsilon}(u)$ as a detached training target; inference remains one generator evaluation.

J. Han, P. Li, Q. Guo, R. Xu, S. Ermon, and E. J. Candès, arXiv:2605.11755, 2026.

W-Flow moves one-step generation toward diffusion-level fidelity



1.29

FID

ImageNet 256×256

$\sim 100\times$

faster sampling

than multi-step diffusion
at similar FID

broader coverage

and domain transfer

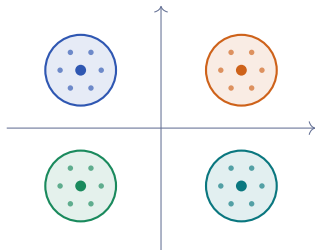
reported improvements over
the compared one-step methods

The result motivates testing Sinkhorn-based one-step generation beyond images.

Results reported by J. Han, P. Li, Q. Guo, R. Xu, S. Ermon, and E. J. Candès, arXiv:2605.11755, 2026.

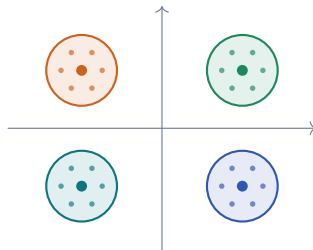
Can we apply W-Flow directly to a wireless channel?

Correct conditional association



each color preserves its transmitted x

Same approximate marginal, wrong channel



conditional clouds are permuted

$p_Y(y) = \int p(y | x)p_X(x) dx$ does not identify $p(y | x)$.

We call the output cloud at fixed x an **output fiber**.

We adapt drifting to the conditional channel law

GLOBECOM 2026
accepted paper

Direct-output
drifting



Residual
drifting

compare output and
residual parameterizations

arXiv / TCOM
journal submission

Direct-output
baseline



Joint
Sinkhorn



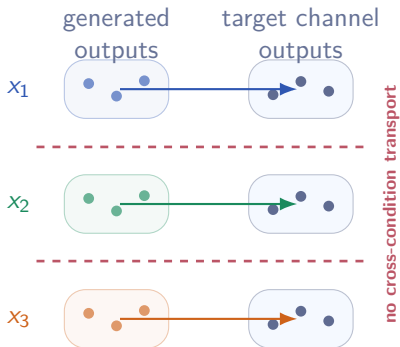
Condition-wise
Sinkhorn

kernel drifting in the selected target
space

Sinkhorn transport in the joint (x, y)
representation

Sinkhorn transport y separately at each
fixed x

Condition-wise Sinkhorn transports only the output law



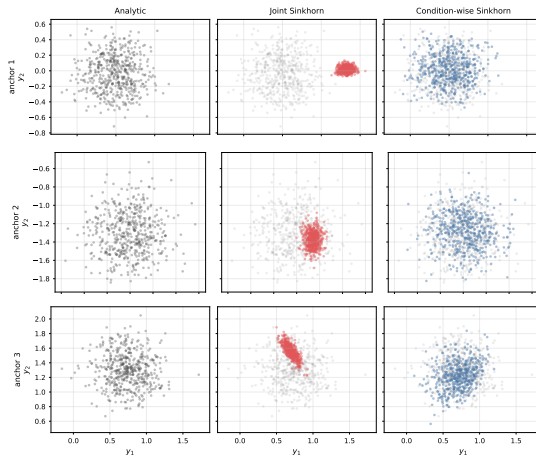
Conditional objective

$$\mathcal{S}_\varepsilon^{\text{cond}}(q_\theta, p) = \int \mathcal{S}_\varepsilon(q_{\theta,x}, p_x) \mu(dx)$$

- ▶ repeated outputs at the same x
- ▶ target coupling gives attraction
- ▶ self coupling removes entropic bias
- ▶ detached regression updates G_θ

R. Fritschek and R. F. Schaefer, "Condition-Wise Sinkhorn Drifting for One-Shot Learned Channel Simulation," arXiv:2606.17893, 2026.

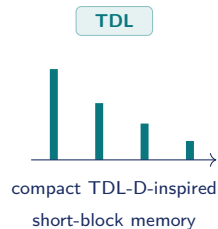
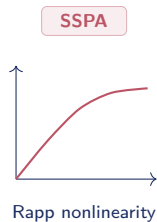
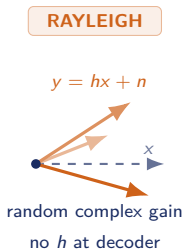
The need for condition-wise Sinkhorn transport



Joint transport displaces or collapses fixed-x clouds.

Condition-wise transport tracks the analytic cloud.

Four channels probe noise, fading, nonlinearity, and memory



AWGN → stochastic fading → nonlinear hardware → structured channel memory

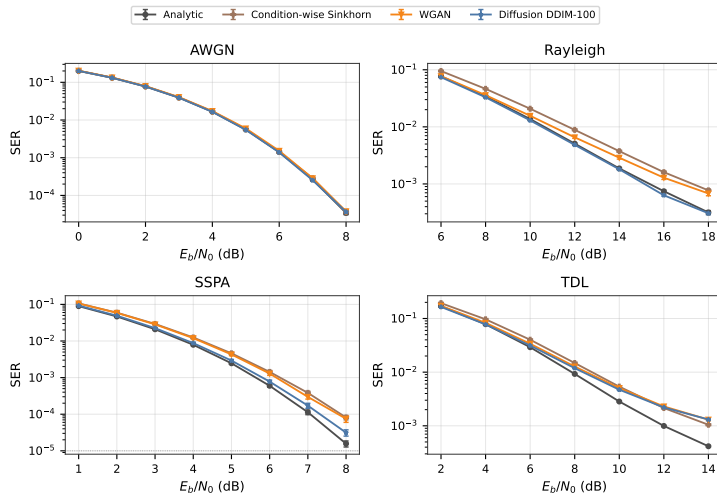
TDL profile: 3GPP TR 38.901, v17.1.0, 2022.

Condition-wise transport improves every anchor-conditioned metric

Channel	Anchor SWD ↓		Anchor GW2 ↓	
	Joint	Condition-wise	Joint	Condition-wise
AWGN	0.1199	0.1154	0.4443	0.4277
Rayleigh	0.1723	0.1097	0.6929	0.3820
SSPA	0.2174	0.0566	0.7876	0.2376
TDL	0.3221	0.1054	1.1871	0.4444

Repeated outputs at fixed transmitted anchors probe the conditional law more directly than global output-space SWD.

Downstream SER reveals both transfer and remaining gaps



One-shot implants use the wall clock for optimization

Optimizer updates in the same 1800 s budget



Same symbolic autoencoder; final evaluation uses the

analytic channel. Mean over 30 seeds.

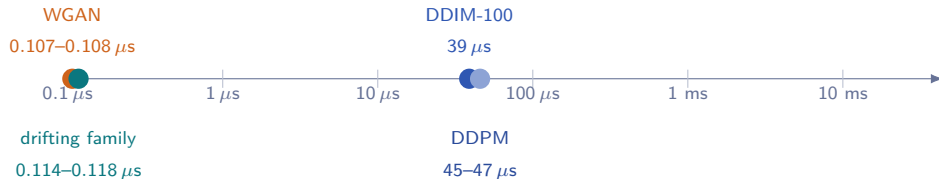
Lowest learned SER under this budget

AWGN	Condition-wise Sinkhorn
Rayleigh	DDIM-100
SSPA	Condition-wise Sinkhorn
TDL	WGAN

The method ranking remains channel-dependent.

The best operating point depends on how the simulator is used

Measured inference time per sample on an NVIDIA RTX 5060 Ti



Standalone fidelity dominates

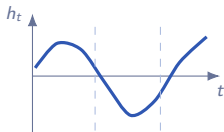
Diffusion can justify sequential sampling on the harder final SER curves.

Repeated channel calls dominate

One-shot generators preserve more time for system optimization and simulation.

Open problems: time variation, nonlinearity, and MIMO

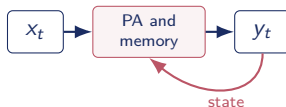
TIME VARIATION



Channels evolve during use

Doppler, nonstationarity, and long temporal memory

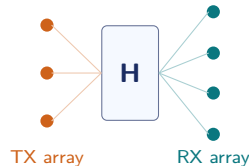
NONLINEARITY



Stateful distortion

Dynamic amplifiers, coupled impairments, and wider operating ranges

SPATIAL STRUCTURE



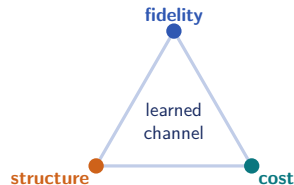
Dimension grows quickly

MIMO correlation, array geometry, and environment state

Goal: preserve $p(y | x)$ as dimension, memory, and channel state grow.

Design channel generators for fidelity, structure, and cost together

1. Generative models provide flexible learned wireless channel simulators.
2. Diffusion offers strong distributional modeling, but iterative sampling can be costly.
3. Conditional drifting explores direct generation without adversarial training.



Conditioning remains critical. Structured, high-dimensional, and measured channels remain open challenges.

Backup: Generic drifting uses attraction and repulsion

For generated particles g_i and target particles p_j , the direct-output field is

$$V(g_i) = \left(\sum_j w_{ij}^P p_j - g_i \right) - \lambda \left(\sum_k w_{ik}^G g_k - g_i \right),$$

where w^P and w^G are row-normalized radial-basis-function kernel weights.

$$\tilde{g}_i = \text{stopgrad}(g_i + \eta V(g_i)), \quad \mathcal{L}_\theta = \frac{1}{B} \sum_i \|f_\theta(z_i) - \tilde{g}_i\|_2^2.$$

Attraction moves particles toward target samples.

Repulsion discourages generated-particle collapse.

Backup: Optimal transport matches probability mass globally

For empirical laws

$$q = \sum_{i=1}^n a_i \delta_{u_i}, \quad p = \sum_{j=1}^m b_j \delta_{v_j}, \quad \sum_i a_i = \sum_j b_j = 1,$$

a coupling is a nonnegative transport matrix with the prescribed marginals:

$$\Pi(a, b) = \left\{ \pi \in \mathbb{R}_+^{n \times m} : \pi \mathbf{1}_m = a, \quad \pi^\top \mathbf{1}_n = b \right\}.$$

Optimal transport selects the least-cost admissible coupling,

$$\text{OT}_c(q, p) = \min_{\pi \in \Pi(a, b)} \sum_{i, j} \pi_{ij} c(u_i, v_j).$$

π_{ij} is the mass transported from u_i to v_j .

For $c(u, v) = \|u - v\|_2^2$, this cost gives $W_2^2(q, p)$.

Backup: Entropic OT produces the W-Flow velocity

Sinkhorn iterations solve the regularized transport problem

$$\text{OT}_\varepsilon(q, p) = \min_{\pi \in \Pi(a, b)} \left\{ \sum_{i, j} \pi_{ij} c(u_i, v_j) + \varepsilon \text{KL}(\pi \parallel a \otimes b) \right\}.$$

Subtracting the two self-costs removes entropic bias:

$$\mathcal{S}_\varepsilon(q, p) = \text{OT}_\varepsilon(q, p) - \frac{1}{2} \text{OT}_\varepsilon(q, q) - \frac{1}{2} \text{OT}_\varepsilon(p, p).$$

The target coupling defines a barycentric transport map,

$$T_{q, p}^\varepsilon(u_i) = \frac{1}{a_i} \sum_j \pi_{ij}^* v_j.$$

Hence the empirical particle velocity is

$$V_\varepsilon(u) = T_{q, p}^\varepsilon(u) - T_{q, q}^\varepsilon(u).$$

Target coupling: attraction toward the data barycenter.

Self coupling: entropic-bias correction and repulsion.

Main experiments: 10 Sinkhorn iterations; minimum $\varepsilon = 10^{-3}$.

Backup: The conditional objective has the desired population equilibrium

With shared input marginal μ ,

$$p(dx, dy) = \mu(dx)p_x(dy), \quad q_\theta(dx, dy) = \mu(dx)q_{\theta,x}(dy).$$

Define

$$\mathcal{S}_\varepsilon^{\text{cond}}(q_\theta, p) = \int \mathcal{S}_\varepsilon(q_{\theta,x}, p_x) \mu(dx).$$

Under the standard Sinkhorn-divergence assumptions,

$$\mathcal{S}_\varepsilon^{\text{cond}}(q_\theta, p) = 0 \quad \Longleftrightarrow \quad q_{\theta,x} = p_x \quad \text{for } \mu\text{-almost every } x.$$

Scope: This is a population statement. Finite minibatch barycentric velocities and detached neural-network regression are practical approximations, especially under model misspecification.

Backup: The estimator uses repeated outputs per transmitted input

For each anchor x_b :

1. Draw $K_g = 4$ generated outputs.
2. Draw $K_p = 4$ target channel outputs.
3. Draw $K_r = 4$ independent generated references.
4. Solve target and self Sinkhorn problems in output space.
5. Regress toward the detached barycentric velocity target.

Leading coupling cost

Condition-wise:

$$O(IBK_g(K_p + K_r))$$

Expanded joint batch:

$$O(IB^2K_g(K_p + K_r))$$

At inference, both variants use one evaluation of the same generator architecture.

Backup: Condition-wise Sinkhorn training update

Input: generator g_θ , channel sampler $p(\cdot \mid x)$, anchors x_i , counts K_g, K_p, K_r , and drift scale η .

Sample at each fixed x_i

1. Generated particles:

$$\hat{y}_{i,k} = g_\theta(x_i, z_{i,k}), \quad k = 1:K_g.$$

2. Positive channel samples:

$$y_{i,j} \sim p(\cdot \mid x_i), \quad j = 1:K_p.$$

3. Independent generated references:

$$\hat{y}'_{i,\ell} = g_\theta(x_i, z'_{i,\ell}), \quad \ell = 1:K_r.$$

Repeat during training. Afterward, sampling is the direct map $\hat{y} = g_\theta(x, z)$.

Transport and project

4. Solve, separately for every i ,

$$\hat{Q}_i \leftrightarrow P_i, \quad \hat{Q}_i \leftrightarrow \hat{Q}'_i.$$

5. Form the barycentric velocity:

$$v_{i,k} = T_{\hat{Q}_i, P_i}^\varepsilon(\hat{y}_{i,k}) - T_{\hat{Q}_i, \hat{Q}'_i}^\varepsilon(\hat{y}_{i,k}).$$

6. Update θ by regressing toward

$$\text{stopgrad}(\hat{y}_{i,k} + \eta v_{i,k}).$$

Backup: Direct-output SWD gives a different ranking

Method	AWGN	Rayleigh	SSPA	TDL
WGAN	.0198 $\pm$.0050	.0171 $\pm$.0050	.0287 $\pm$.0156	.0289 $\pm$.0063
DDPM	.0070 $\pm$.0004	.0079 $\pm$.0008	.0032 $\pm$.0005	.0297 $\pm$.0011
DDIM-100	.0042 $\pm$.0006	.0044 $\pm$.0007	.0024 $\pm$.0004	.0248 $\pm$.0009
Direct drifting	.0100 $\pm$.0007	.0085 $\pm$.0012	.0060 $\pm$.0008	–
Joint Sinkhorn	.0148 $\pm$.0002	.0152 $\pm$.0002	.0117 $\pm$.0002	.0966 $\pm$.0005
Condition-wise Sinkhorn	.0058 $\pm$.0001	.0073 $\pm$.0001	.0071 $\pm$.0002	.0118 $\pm$.0002

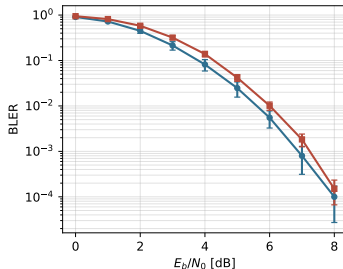
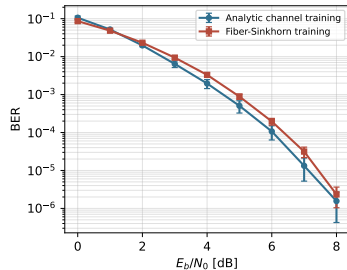
Reference WGAN/diffusion rows use their stable benchmark protocols. The drifting-family block is the controlled matched-architecture comparison; bold marks its best value. Reference runs use 10 seeds, W-Flow rows use 100 seeds, and the SSPA condition-wise row uses the 30-seed update-budget-controlled run.

Backup: SSPA requires an update-budget-controlled operating point

Run	Updates	Seeds	Direct SWD	Anchor-SWD ratio	Anchor-GW2 ratio	SER at 8 dB
Short local	0.31k	1	1.07×10^{-1}	5.14	4.10	–
Early compact	0.88k	1	4.09×10^{-2}	4.50	3.88	–
Reported point	4.69k	30	7.07×10^{-3}	1.13	1.15	9.33×10^{-5}
Full preset	390.6k	100	1.53	35.5	60.1	2.70×10^{-2}

Ratios divide learned-versus-analytic anchor metrics by an analytic-versus-analytic floor from independent channel draws. The sharp same-condition field converges early; with fixed drift scale and small same-condition sample sets, the long diffusion-style preset is a poor long-horizon optimizer. The full-preset row is a sensitivity check and is not used for model selection.

Backup: The AWGN result transfers to block length 64



TurboAE-style check

- ▶ block length 64
- ▶ rate 1/2
- ▶ training at 4 dB
- ▶ 30 seeds
- ▶ analytic evaluation

BER at 4 dB

- ▶ analytic: 1.96×10^{-3}
- ▶ condition-wise: 3.30×10^{-3}

Backup: Scope and reproducibility boundaries

Controlled conclusions

- ▶ W-Flow variants share the same one-shot generator.
- ▶ Drift-field differences drive their ablation ranking.
- ▶ Cross-family baselines use stable reference protocols.
- ▶ Comparisons are contextual, not fully matched capacity.

Current limitations

- ▶ exact estimator assumes repeated outputs at fixed x
- ▶ local measured-data approximation is not yet validated
- ▶ compact TDL is not a full OFDM-scale evaluation
- ▶ global SWD must be paired with conditional and downstream metrics

Timing protocol

PyTorch float32 on NVIDIA RTX 5060 Ti; batch size 512; CUDA synchronization; no graph capture, compilation, or mixed precision. Source code:
github.com/Fritschek/conditional_drifting_models_public

Backup: Inference-network parameter counts

Model	AWGN	Rayleigh	SSPA	TDL
Direct drifting	20,487	20,487	20,744	20,744
Sinkhorn drifting variants	20,487	20,487	20,744	20,744
Diffusion denoiser	59,847	74,247	60,178	74,632
WGAN generator	19,335	71,431	72,200	72,200

Sinkhorn couplings are training-time computations and add no inference-time parameters. DDPM and DDIM share denoiser weights. Cross-family architectures and training schedules follow stable family-specific protocols rather than a fully matched-capacity design.

Backup: Selected references

- ▶ I. Goodfellow et al., “Generative Adversarial Nets,” NeurIPS, 2014.
- ▶ M. Arjovsky, S. Chintala, and L. Bottou, “Wasserstein Generative Adversarial Networks,” ICML, 2017.
- ▶ J. Ho, A. Jain, and P. Abbeel, “Denoising Diffusion Probabilistic Models,” NeurIPS, 2020.
- ▶ P. Dhariwal and A. Nichol, “Diffusion Models Beat GANs on Image Synthesis,” NeurIPS, 2021.
- ▶ J. Song, C. Meng, and S. Ermon, “Denoising Diffusion Implicit Models,” ICLR, 2021.
- ▶ R. Rombach et al., “High-Resolution Image Synthesis with Latent Diffusion Models,” CVPR, 2022.
- ▶ M. Kim, R. Fritschek, and R. F. Schaefer, “Diffusion Models for Accurate Channel Distribution Generation,” accepted, IEEE TMLCN, 2026.
- ▶ M. Kim, R. Fritschek, and R. F. Schaefer, “Robust Generation of Channel Distributions with Diffusion Models,” IEEE ICC, 2024.
- ▶ M. Deng, H. Li, T. Li, Y. Du, and K. He, “Generative Modeling via Drifting,” arXiv:2602.04770, 2026.
- ▶ J. Han et al., “One-Step Generative Modeling via Wasserstein Gradient Flows,” arXiv:2605.11755, 2026.
- ▶ R. Fritschek and R. F. Schaefer, “Condition-Wise Sinkhorn Drifting for One-Shot Learned Channel Simulation,” arXiv:2606.17893, 2026.
- ▶ M. Cuturi, “Sinkhorn Distances: Lightspeed Computation of Optimal Transport,” NeurIPS, 2013.
- ▶ J. Feydy et al., “Interpolating between Optimal Transport and MMD using Sinkhorn Divergences,” AISTATS, 2019.